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Ternary liquid–liquid equilibria of bis(trifluoromethylsulfonyl)-amide based ionic liquids + thiophene + *n*-heptane. The influence of cation structure

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ABSTRACT

Ternary liquid–liquid equilibria for 3 systems containing ionic liquids (4-(2-methoxyethyl)-4-methylmorpholinium bis(trifluoromethylsulfonyl)-amide, 1-(2-methoxyethyl)-1-methylpiperidinium bis-(trifluoromethylsulfonyl)-amide, 1-(2-methoxyethyl)-1-methylpyrrolidinium bis(trifluoromethylsulfonyl)-amide) + thiophene + *n*-heptane have been determined at T = 298.15 K. All systems showed high solubility of thiophene in the ionic liquid and low solubility of *n*-heptane. The solute distribution coefficient and the selectivity were calculated for all systems. High selectivities were obtained. The experimental results have been correlated using NRTL model. The influence of cation structure on phase equilibria is discussed.

1. Introduction

Ionic liquids (ILs) have become one of the most studied classes of compounds since the late 1990s. They are salts melting at relatively low temperatures due to asymmetry in structure. Because of their extremely low volatility, high thermal and chemical stability and wide liquid range, ILs are considered as good potential "green" replacements for conventional volatile and often toxic organic solvents in many areas of academic and industrial chemistry. Almost unlimited combination of anions and cations allows tailoring properties of ionic liquids which make them "designer solvents", therefore the knowledge about influence of IL structure on physicochemical properties is very important. Thiophene forms with *n*-heptane azeotropic mixture at 0.832 mass fraction of thiophene at T = 356.3 K [1], therefore the separation of this mixture is not possible using rectification. Alternative method, such as liquid–liquid extraction can be used, but in this case the knowledge about ternary liquid–liquid equilibria is needed. Previous works indicate that ionic liquids can be used in separation of thiophene–alkane mixtures [2–6], as well as other mixtures [7–9]. Additionally ionic liquids are considered as extractants in separation of sulfur compounds from fuels [10–12], therefore provided data will be useful in this area of interest.

2. Experimental

2.1. Chemicals

The list of ionic liquids used in this study including source and grade is as follows: $[COC_2mMOR][NTf_2]$, MERCK, grade >0.995 mass fraction; $[COC_2mPIP][NTf_2]$, MERCK, grade >0.995 mass fraction; $[COC_2mPYR][NTf_2]$, MERCK, grade >0.995 mass fraction; The ionic liquids were further purified by subjecting the liquid to a very low pressure of about 5×10^{-3} Pa at a temperature about 95 °C for *ca.* 5 h. This procedure removed any volatile chemicals and water from the ionic liquid. Thiophene had a purity of >0.9988 mass fraction (GC analysis) and was supplied by Fluka. *n*-Heptane had a purity of >0.9979 mass fraction (GC analysis) and was supplied by Fluka. Acetone had a purity of >0.9997 mass fraction (GC analysis) and was supplied by Fluka. 1-Propanol had a purity of >0.9987 mass fraction (GC analysis) and was supplied by Sigma-Aldrich. Structures of investigated ionic liquids are presented in Table 1.

2.2. Apparatus and procedure

The water content of ILs was analyzed by Karl–Fischer titration technique (model SCHOTT Instruments TitroLine KF). A sample of IL was dissolved in methanol and titrated with steps of 2.5 μL. The results obtained have shown the water content to be less than 200 ppm, 180 ppm and 160 ppm for $[COC_2mMOR][NTf_2]$, $[COC_2mPIP][NTf_2]$, and $[COC_2mPYR][NTf_2]$, respectively. The titrant CombiTitrant 5 no. 1.8805.1000 Merck one-component reagent for volumetric Karl Fischer titration was

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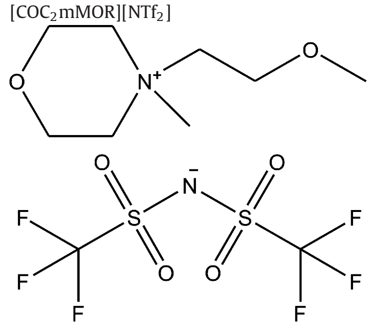
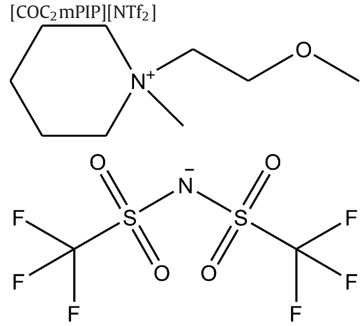
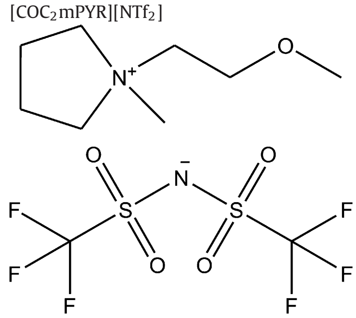
Table 1
Structures, decomposition temperature, T_d, density, ρ and viscosity, η, as a function of temperature for investigated ILs.[a]

Structure	T/K	ρ/g cm^{-3}	η/mPa s
[COC$_2$mMOR][NTf$_2$]  T_d/K = 680	298.15	1.50040[b]	306.1
	308.15	1.49091[b]	155.9
	318.15	1.48181[b]	90.68
	328.15	1.47270[b]	57.23
	338.15	1.46361[b]	38.69
	348.15	1.45457[b]	27.54
[COC$_2$mPIP][NTf$_2$]  T_d/K = 702	298.15	1.43494	102.6
	308.15	1.42587	63.22
	318.15	1.41683	41.92
	328.15	1.40785	29.43
	338.15	1.39891	21.53
	348.15	1.39005	16.32
[COC$_2$mPYR][NTf$_2$]  T_d/K = 708	298.15	1.45384	54.52
	308.15	1.44446	36.94
	318.15	1.43513	26.41
	328.15	1.42587	18.23
	338.15	1.41667	14.06
	348.15	1.40754	11.07

[a] Standard uncertainties u are $u(T)$ = 0.01 K and 0.05 K for ρ and η, respectively, $u(\rho) = 10^{-5}$ g cm^{-3}, $u(\eta) < 0.1\%$.
[b] From Ref. [16].

used. The lower determination limit of this technique is approximately 50–100 ppm H_2O. Densities were measured using an Anton Paar GmbH 4500 vibrating-tube densimeter (Graz, Austria) with uncertainty of 10^{-5} g cm^{-3}. Viscosity measurements were performed using an Anton Paar GmbH AMVn Automated Micro Viscometer (Graz, Austria) with uncertainty of 0.1%. Results are presented in Table 1. The decomposition temperature of ILs has been measured using thermogravimetric analysis (TGA) performed using TA Instruments Q600 thermogravimetric analyzer. Results are presented in Table 1 and Figs. 1S–3S, Supplementary material. Detailed description of above mentioned methods can be found in previous work [13].

The experimental LLE tie-lines were determined by preparation of immiscible mixtures of the three compounds of the ternary system studied. Mixtures were placed into a jacketed glass cell with a volume of 10 cm^3, together with a coated magnetic stirring bar, and closed to avoid losses of substances by evaporation or absorb of moisture from the atmosphere. The jackets were connected to a thermostat bath (LAUDA Alpha) to maintain a constant temperature of 298.15 ± 0.02 K in the vessels. The mixtures were stirred using the magnetic stirrer for 1 h to reach the thermodynamic equilibrium and then allowed to settle for a minimum of 20 h to guarantee that the equilibrium state was completely reached. After phase separation, approximately 0.1–0.3 cm^3 samples from both phases in equilibrium (a denser ionic liquid-rich phase at the bottom, and a less dense hydrocarbon-rich phase on top) were taken using glass syringes with stainless steel needles, without disturbance of the interface. Acetone (0.5 cm^3) was added to the samples to avoid phase splitting and to maintain a homogeneous mixture. 1-Propanol was used as internal standard for the GC-analysis. Since ionic liquids have either no or a very low vapor pressure, they cannot be analyzed by GC, therefore, only thiophene and n-heptane were analyzed. For a ternary mixture, only two components need to be analyzed, therefore the ionic liquid composition was calculated by difference.

The composition was analyzed by gas chromatography. The PerkinElmer Clarus 580 GC equipped with FID detector was used. The data were collected and processed using TotalChrom Workstation software. The parameters of analysis were as follow: column: PerkinElmer Elite-5 (5% diphenyl–95% dimethyl polysiloxane), 30 m length, 0.53 mm internal diameter, 5 μm film thickness; column oven temperature: 70 °C for 10 min; carrier gas: helium; flow rate: 5 ml min^{-1}; injector temperature: 220 °C; detector temperature: 220 °C; split ratio: 5:1. To avoid of column contamination the ionic liquid was collected in the injector liner (filled with quartz glass) and in the pre-column. All samples were injected three times, and the average value was calculated. The uncertainty of the mole fraction did not exceed ±0.002.

3. Modeling

The LLE was correlated with the NRTL model describing the excess Gibbs energy [14]. The equations were described by us earlier [8,15]. The NRTL nonrandom parameter α was set to a value of $\alpha = 0.3$, which has given the best results of the correlation. The results of the correlation, values of the model parameters, and the corresponding standard deviations are given in Table 3. The results of the correlation are presented in Figs. 1–3.

4. Results and discussion

The compositions of the experimental tie-lines for the ternary systems IL + thiophene + *n*-heptane at $T = 298.15$ K are reported in Table 2. The corresponding triangular diagrams with the experimental tie-lines for each system are shown in Figs. 1–3.

Values of selectivity S and solute distribution ratio β reported in Table 2 are calculated from experimental data according to equations:

$$S = \frac{x_2^{\mathrm{IL}} x_3^{\mathrm{HC}}}{x_2^{\mathrm{HC}} x_3^{\mathrm{IL}}} \quad (1)$$

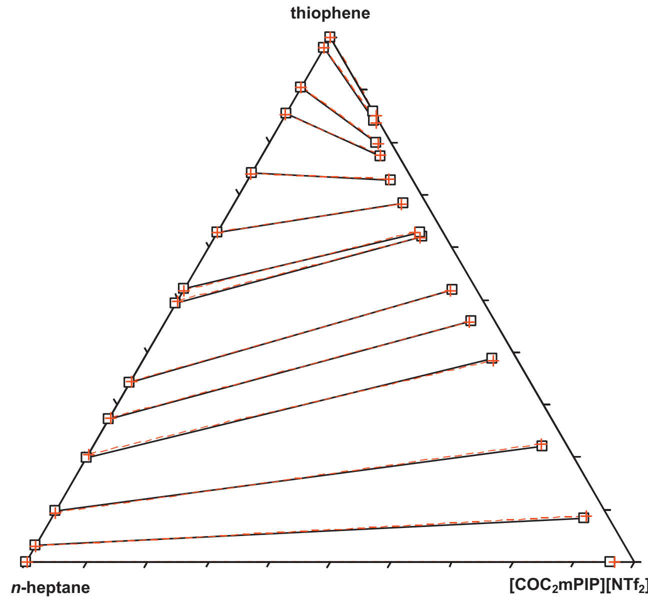


Fig. 2. Experimental *versus* NRTL correlations for the composition tie lines of the ternary system {[COC$_2$mPIP][NTf$_2$] (1) + thiophene (2) + *n*-heptane (3)} at 298.15 K; (□—□) experimental data, (+—+) NRTL correlation.

$$\beta = \frac{x_2^{\mathrm{IL}}}{x_2^{\mathrm{HC}}} \quad (2)$$

where x is the mole fraction, subscripts 2 and 3 refer to thiophene and to *n*-heptane, respectively, and superscripts HC and IL indicate the hydrocarbon-rich phase and the IL-rich phase, respectively.

The value of selectivity is related to the number of stages in the separation process. From the investigated ionic liquids the highest values of selectivities are for [COC$_2$mMOR][NTf$_2$] (Fig. 4). Ionic liquids based on piperidinium and pyrrolidinium cations, [COC$_2$mPIP][NTf$_2$] and [COC$_2$mPYR][NTf$_2$], reveal very similar selectivities due to the small difference in the structures.

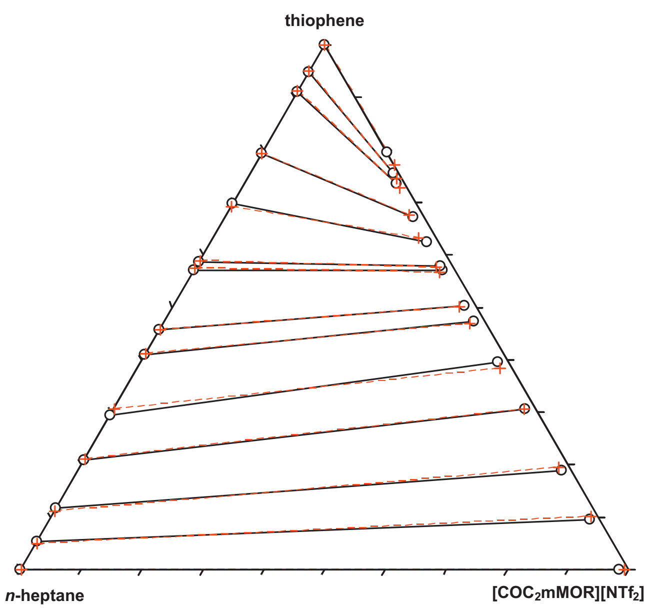


Fig. 1. Experimental *versus* NRTL correlations for the composition tie lines of the ternary system {[COC$_2$mMOR][NTf$_2$] (1) + thiophene (2) + *n*-heptane (3)} at 298.15 K; (○—○) experimental data, (+—+) NRTL correlation.

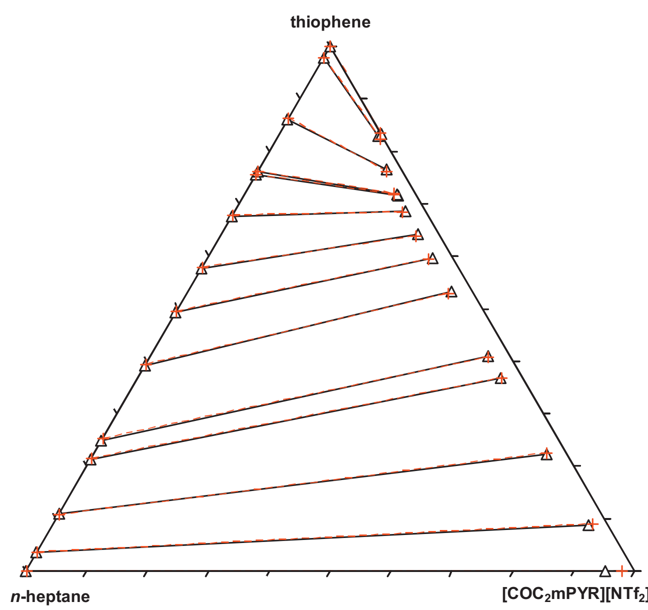


Fig. 3. Experimental *versus* NRTL correlations for the composition tie lines of the ternary system {[COC$_2$mPYR][NTf$_2$] (1) + thiophene (2) + *n*-heptane (3)} at 298.15 K; (Δ—Δ) experimental data, (+—+) NRTL correlation.

Table 2
Compositions of experimental tie lines, solute distribution ratios, β and selectivity, S for ternary systems {IL (1) + thiophene (2) + n-heptane (3)} at T = 298.15 K.[a]

Hydrocarbon-rich phase		IL-rich phase		S	β
x_1^{HC}	x_2^{HC}	x_1^{IL}	x_2^{IL}		
$[COC_2mMOR][NTf_2]$					
0.000	0.000	0.984	0.000		
0.000	0.054	0.888	0.096	104.3	1.76
0.000	0.118	0.795	0.189	88.4	1.60
0.000	0.209	0.676	0.307	70.4	1.47
0.000	0.295	0.588	0.395	54.7	1.34
0.000	0.409	0.509	0.473	38.1	1.16
0.000	0.457	0.479	0.503	33.1	1.10
0.000	0.571	0.409	0.571	21.8	1.00
0.000	0.587	0.401	0.578	20.0	0.98
0.000	0.697	0.357	0.624	14.2	0.89
0.000	0.793	0.309	0.673	9.9	0.85
0.000	0.911	0.251	0.736	5.2	0.81
0.000	0.949	0.235	0.756	4.6	0.80
0.000	1.000	0.204	0.796		0.80
$[COC_2mPIP][NTf_2]$					
0.000	0.000	0.961	0.000		
0.000	0.032	0.876	0.084	62.9	2.64
0.000	0.097	0.739	0.220	49.8	2.27
0.000	0.199	0.572	0.387	38.0	1.94
0.000	0.273	0.501	0.460	31.4	1.69
0.000	0.342	0.441	0.519	25.4	1.52
0.000	0.493	0.340	0.621	16.6	1.26
0.000	0.521	0.334	0.629	15.3	1.21
0.000	0.628	0.277	0.684	10.6	1.09
0.000	0.741	0.236	0.727	6.8	0.98
0.000	0.855	0.195	0.774	4.2	0.90
0.000	0.904	0.176	0.799	3.4	0.88
0.000	0.980	0.152	0.841	2.3	0.86
0.000	1.000	0.142	0.858		0.86
$[COC_2mPYR][NTf_2]$					
0.000	0.000	0.951	0.000		
0.000	0.036	0.880	0.087	70.6	2.39
0.000	0.109	0.744	0.223	54.8	2.04
0.000	0.212	0.596	0.368	38.5	1.74
0.000	0.249	0.555	0.410	34.8	1.65
0.000	0.392	0.434	0.532	24.4	1.36
0.000	0.493	0.371	0.596	18.4	1.21
0.000	0.577	0.324	0.642	13.7	1.11
0.000	0.677	0.280	0.687	9.8	1.01
0.000	0.756	0.253	0.716	7.5	0.95
0.000	0.762	0.252	0.717	7.3	0.94
0.000	0.861	0.210	0.764	4.8	0.89
0.000	0.978	0.164	0.830	2.7	0.85
0.000	1.000	0.167	0.833		0.83

[a] Standard uncertainties u are $u(x) < 0.002$, $u(T) = 0.02$ K.

Substitution of —CH_2— group with oxygen in piperidinium structure (giving morpholinium cation) causes almost double increase of selectivities. In each case the selectivities decrease for higher solute concentrations.

Table 3
Values of binary parameters for the NRTL equation[a] for the ternary systems {IL (1) + thiophene (2) + n-heptane (3)} at T = 298.15 K.

IL	ij	g_{ij}/J mol^{-1}	g_{ji}/J mol^{-1}	σ_x
$[COC_2mMOR][NTf_2]$	12	−2949.6	24,811	0.0058
	13	9907.4	12,183	
	23	4109.9	−2230.3	
$[COC_2mPIP][NTf_2]$	12	−2207.5	33,568	0.0031
	13	6626.6	12,221	
	23	4023.3	−1123.6	
$[COC_2mPYR][NTf_2]$	12	−1774.9	32,877	0.0034
	13	7423.7	11,720	
	23	3935.5	−1103.1	

[a] Parameter α = 0.3.

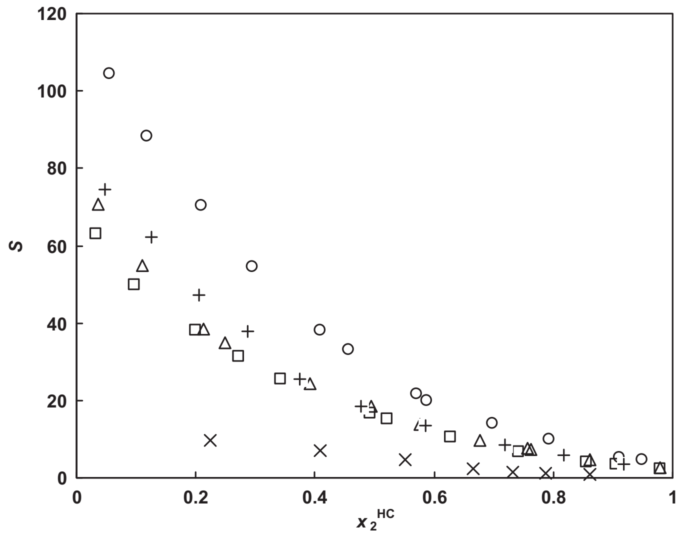


Fig. 4. Selectivity, S as a function of the mole fraction of thiophene in the hydrocarbon-rich phase for the investigated systems; (○) $[COC_2mMOR][NTf_2]$, (□) $[COC_2mPIP][NTf_2]$, (△) $[COC_2mPYR][NTf_2]$, (+) $[emim][NTf_2]$ [3], (×) $[omim][NTf_2]$ [4].

The solute distribution ratios indicate the solute-carrying capacity of the investigated ionic liquid. For effective extraction values of β higher than 1 are desired. Fig. 5 shows the solute distribution ratios for three investigated ionic liquids as a function of the thiophene molar fraction in the hydrocarbon-rich phase. The highest values of β are for ionic liquids based on piperidinium and pyrrolidinium cations, above 2 in the region of low concentration of thiophene and below 2 for morpholinium-based ionic liquid. Obtained results suggest that investigated ionic liquids could efficiently perform the separation of thiophene from n-heptane, especially $[COC_2mMOR][NTf_2]$.

Obtained data are related to the literature values for other ionic liquids based on $[NTf_2]^-$ anion, 1-ethyl-3-methyl-imidazolium bis(trifluoromethylsulfonyl)-amide, $[emim][NTf_2]$ [3] and 1-octyl-3-methyl-imidazolium bis(trifluoromethylsulfonyl)-amide, $[omim][NTf_2]$ [4]. Ionic liquid with more aliphatic character

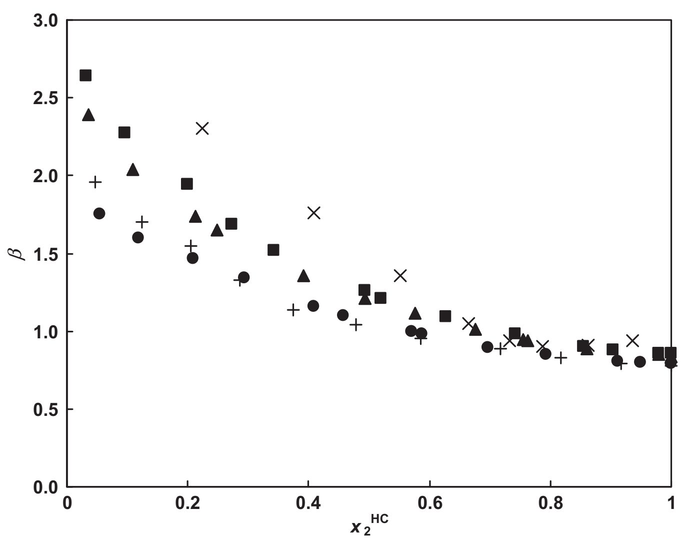


Fig. 5. Solute distribution ratio, β as a function of the mole fraction of thiophene in the hydrocarbon-rich phase for the investigated systems; (●) $[COC_2mMOR][NTf_2]$, (■) $[COC_2mPIP][NTf_2]$, (▲) $[COC_2mPYR][NTf_2]$, (+) $[emim][NTf_2]$ [3], (×) $[omim][NTf_2]$ [4].

like [omim][NTf_2] has the lowest values of selectivities, but in this case the β values are the highest, opposite to [emim][NTf_2]. Comparing to the ILs from this work, [emim][NTf_2] has selectivities slightly larger than [COC_2mPIP][NTf_2] and [COC_2mPYR][NTf_2] but solute distribution ratios are much lower, comparable to [COC_2mMOR][NTf_2].

5. Conclusions

An experimental investigation on liquid–liquid equilibria of the ternary mixtures composed of tree ionic liquids based on bis(trifluoromethylsulfonyl)-amide anion + thiophene + *n*-heptane has been carried out at T = 298.15 K. It was shown that investigated ionic liquids could be used as entrainer in separation of thiophene from *n*-heptane, especially 4-(2-methoxyethyl)-4-methylmorpholinium bis(trifluoromethylsulfonyl)-amide. The results were correlated with the NRTL equation providing an excellent fit of the experimental data.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.fluid.2012.02.019.

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